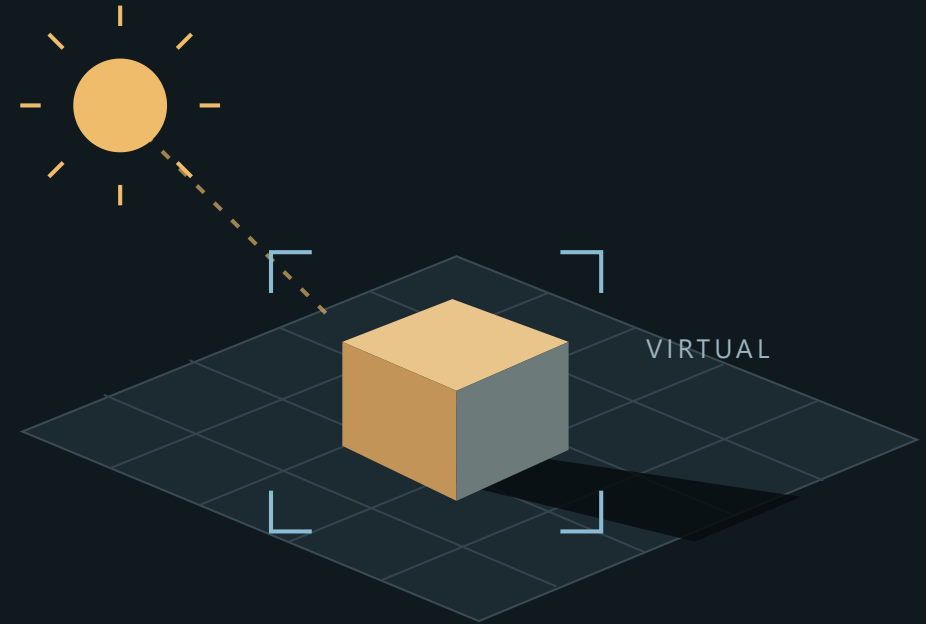


Sun-Driven Lighting for Location-Based WebXR

Making virtual sunlight
match the real world

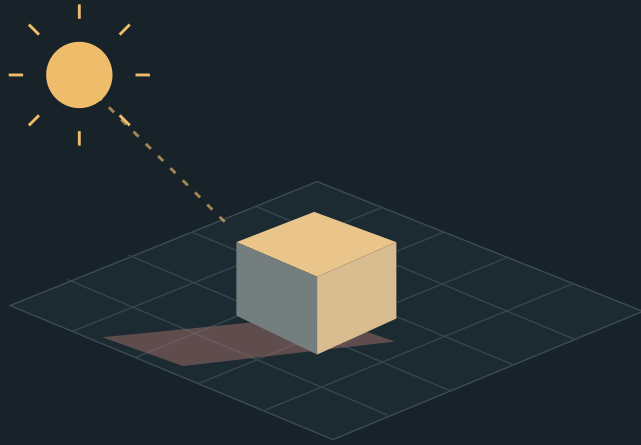
Alissa Farber · Chin-Lucas Woo



Project Goal

FIXED LIGHTING

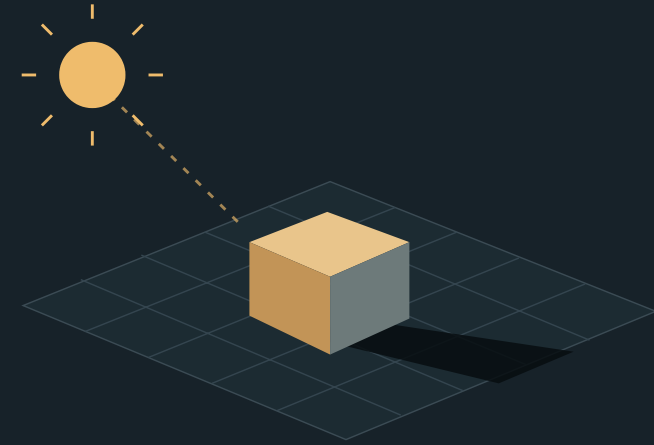
MISMATCH



Correct position, **wrong lighting**

SUN-DRIVEN LIGHTING

INTENDED



Real location + time →
realistic lighting

Can virtual lighting match the real environment?

Task 1: Recorder Testing

TASK

Walk. Revisit. Record.

Outdoor recorder walks

Repeated reference points

Observe heading + lighting/shadows

Collect data for replay validation

OUR OBSERVATIONS

GPS + heading were stable.

GPS

Most accurate

AR

Slightly less accurate

Fused

Less accurate

No useful lighting/shadows visible.

Required ZIP not generated on Android.



Our Component Architecture



PURE LOGIC

Sun position + lighting model



VISUALIZATION COMPONENTS

Sun disc + shadow rig



FRAMEWORK / AR INTEGRATION

Adapter connects inputs; scene composes outputs

Splitting the feature into independent components made parallel development easier and kept integration manageable.

Sun Position Core

Latitude

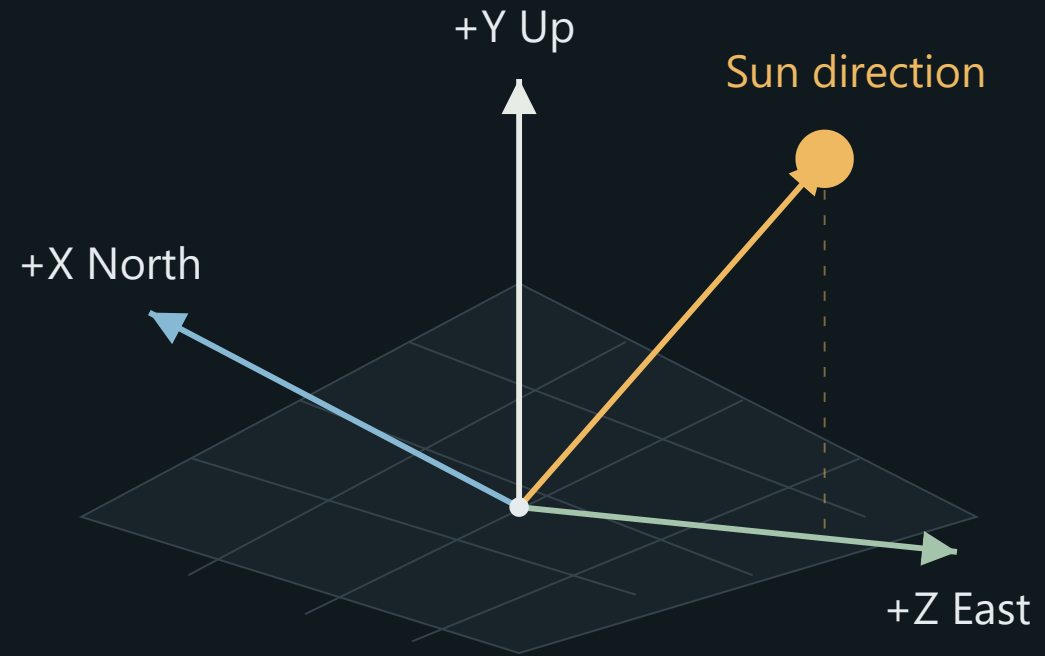
Longitude

UTC Time



Sun Position

Correct sun angles still need
the correct coordinate
conversion.



SunCalc convention



NUE conversion



Normalized direction

[LIVE DEMO](#) Sun Position Demo

A Coordinate-System Mistake

AI-GENERATED PLAN

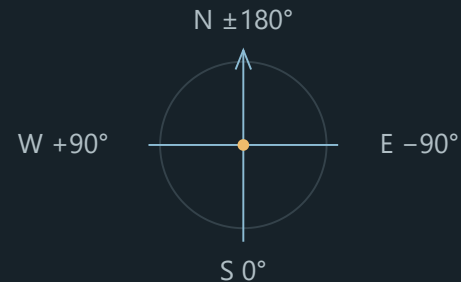
"Azimuth measured clockwise from North"



Mismatch found during review

SUNCALC ACTUALLY USES

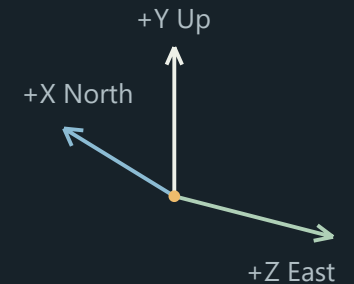
0°	South
+90°	West
-90°	East
±180°	North



Correct conversion

NUE

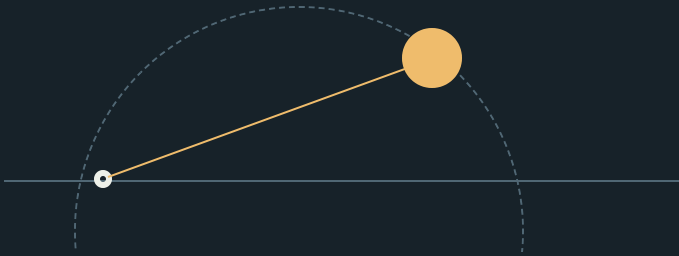
+X North
+Y Up
+Z East



Takeaway: A plausible assumption can still be wrong.

Visual Components

One calculated sun state drives all visual effects.



Visible Sun Disc

INPUT

Sun direction

OUTPUT

Visible sun position

SECOND PURPOSE

AR alignment check

LIVE DEMO

[Visible Sun Disc Demo](#)



Lighting Model

INPUT

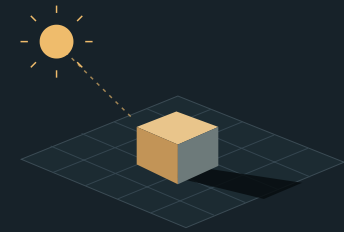
Sun altitude

OUTPUT

Light intensity + color

LIVE DEMO

[Sun Altitude Lighting Demo](#)



Shadow Rig

INPUT

Sun direction + lighting values

OUTPUT

Directional light + matching virtual shadows

LIVE DEMO

[Shadow Rig Demo](#)

Real Data Adapter

REAL USE / PRODUCTION

📍 Live GPS

+

🕒 Current time



Real Data Adapter



Sun position

TESTS / DEMOS

📍 Fixed location

+

🕒 Fixed time



Real Data Adapter



Repeatable result

AVAILABLE MODES

| Live GPS + Real Time

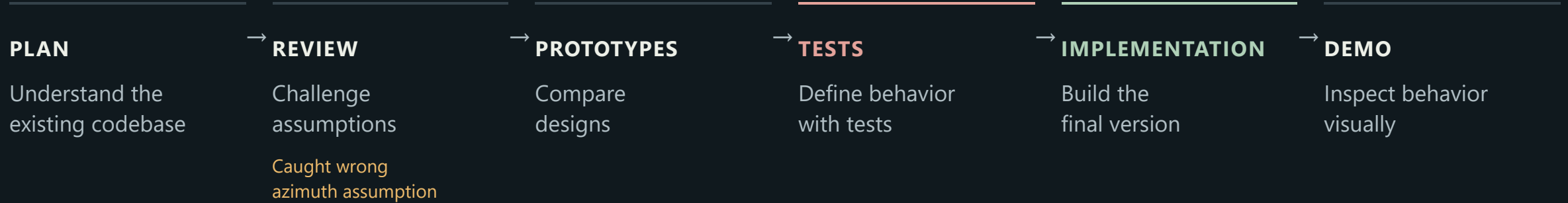
| Live GPS + Fixed Time

| Fixed Location + Real Time

| Fixed Location + Fixed Time

LIVE DEMO Real Data Adapter Demo

Our AI-Assisted Workflow



REAL DATA ADAPTER / THREE APPROACHES



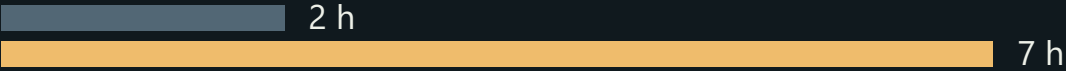
The simpler approach became the starting point.
AI supported planning, review and implementation.

Time Estimations

Estimated Actual

Sun Position

SunCalc + NUE conversion + unit tests + demo



Sun Disc

Unit tests + demo



Real Data Adapter

Matched estimate



Lighting Model



Shadow Rig

Matched estimate



Task 1

Recorder workflow + no ZIP export



Largest underestimate

Sun Position: 2 h → 7 h

On estimate

Adapter + Shadow Rig

Final Integration

GPS + Time



Real Data Adapter



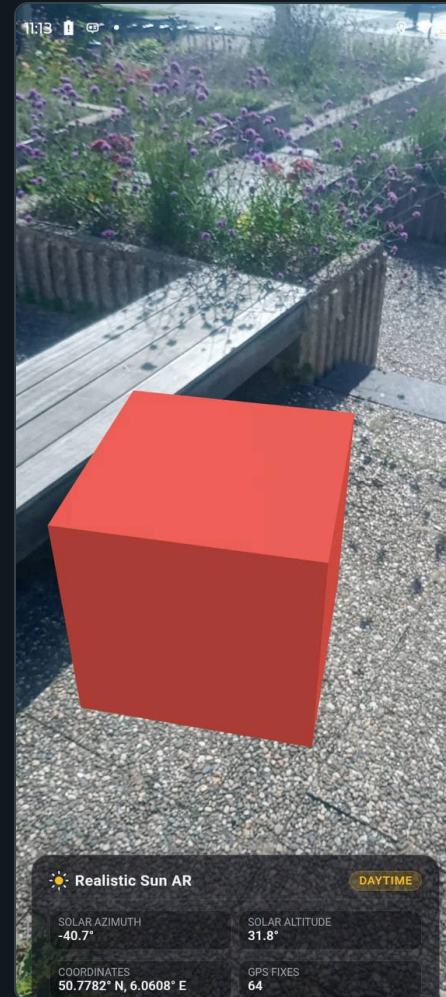
Sun Position



Sun Disc + Lighting
+ Shadows



AR Scene



Outdoor test of the integrated AR scene

What We Learned

WHAT WE LEARNED

- Small technical details can have a large impact
- Testing is essential for real-world data
- AI output requires human review
- Modular development makes parallel work easier

WHAT WE WOULD CHANGE

- Validate external tools earlier
- Plan more time for testing and integration
- Work more closely together during integration

Thank you.

Any questions?